Surname and Initials:	Session Day:	Group no
Reyneke, R	Wednesday	6
Student number: 21495034		

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

DEPARTMENT OF PHYSICS

FSK 116

Practical Report

PAR – Projectile Motion

FSK 116 Online Practical Activity: Projectile motion

Total: 40 marks.

Instructions:

The data provided in the practical, is read data captured by means of the *Tracker* programme from a video I made at home using my cell phone. The idea is that you use this data, but if you are more adventurous, download the programme from https://physlets.org/tracker/ (free, open source) and make your own.

Follow the instructions below and upload your answers as a PDF with the answers clearly numbered.

Apparatus

Optional: Tracker (downloadable from https://physlets.org/tracker/ or, using free data via https://connect.up.ac.za/https/drive.google.com/file/d/1z2P3W99UcyrTD3WXwvy8P8bXyCSthJvs/view?usp=sharing) installed on a computer, a cell phone, dark coloured ball, a well-lit white wall, four markers with Prestik, a tape measure. Graph paper or spreadsheet such as Excel or Google Sheets (Google sheets is available online using free data as https://connect.up.ac.za/https/docs.google.com/spreadsheets/) If this link does not work, try copying it and pasting it in directly on your browser or log into Gmail from the connect.up.ac.za portal, and then go to "Google Apps" and select "Sheets")

The video is available online as:

Direct: https://drive.google.com/file/d/1Q3wNT42ZVv5jmikNe0ErmmXhermWvh2-/view?usp=sharing
Through connect: https://connect.up.ac.za/https/drive.google.com/file/d/1Q3wNT42ZVv5jmikNe0ErmmXhermWvh2-/view?usp=sharing

The data as a spreadsheet is available as:

Direct: https://drive.google.com/file/d/1u1NTBhv0JGdyyJC1lA832eZEIPJFINb-/view?usp=sharing
Through connect: https://connect.up.ac.za/drive.google.com/file/d/1u1NTBhv0JGdyyJC1lA832eZEIPJFINb-/view?usp=sharing

PART A: Capturing the data (optional, no marks, but a lot of fun ☺)

[A1] Optional, but fun: I did this from scratch with no experience, and it took two hours and was a lot of fun. Writing the instructions up took longer ☺. Note: Tracker tries its best to be user friendly – read the hints it gives on the screen.

a) Download and install the Tracker software.
b) Pick an evenly and well-lit, white wall. (I had a plant and some shadows from the washing line in the way. It is not ideal, but one can work around it.)
c) Find a place where you can launch the ball from so that the ball travels in front of the wall for a considerable part of its trajectory. Ideally, it has to reach its maximum height and drop down to the level that you released if from.
d) For calibration make two (removable!) marks on the wall a fixed horizontal distance apart. I stuck plastic wheels to the wall using Prestik. Do the same for the vertical calibration. (I picked 1 m for convenience.)
e) Place your cell phone at a point where the whole motion can be recorded, preferably facing the wall perpendicularly.
f) Record a video of the ball flying. (If possible, set the shutter speed to as short as possible – on some phones it means setting the ISO number high, using a high frame rate (30 fps) and a high resolution.)
g) Open Tracker. Click on "Video"->"Import" and import your video.
h) You probably recorded far too much, so use the small black triangles beneath the timeline to select the region of interest from the frame showing you releasing the ball to the frame just before the ball hits the ground. (You can go forward or backwards in steps of one frame at a time).
i) If your video has interference like plants on it, right click and select "Filters"->"New"->"Negative" and "Filters"->"New"->"Baseline" (capture a frame without the ball) to subtract everything leaving you with just be ball (and in my case, its shadow) in white moving across a black screen.
j) Click on "Track"->"New"->"Point Mass"
k) Go to frame 1, and Ctrl-Shift-Click on the ball to mark it. Zoom in to define the template of the ball clearly (circle) and the search area (rectangle) to include the box.
l) On the autotracker, click "Search This" to mark and select the ball.
m) On the autotracker, click "Search Next". If all went well, Tracker will have found the ball on the next frame and marked it (there will be a report in green text). If not, there will be a report in red text. Help autotracker to find the ball by moving the search are, or redefining the template (Ctrl-Shift-Click). If all works well, Tracker should track all the frames with maybe one or two exception.
n) Repeat clicking "Search Next" until the ball is found on all frames. (The success of this depends on the quality of your video. I had plants in the way so I selected the filters "invert" and "baseline" to subtract everything that was not moving. Play around with the search area and evolution rate to make it happen. Note: You can zoom in to make it easier.)
o) Select "Track"->"New"->"Coordinate System". Move the coordinate system so that its origin is at the point where the ball is released, and the axes are horizontal and vertical.

p) Select "Track"->"New"->"Calibration Tools"->"Calibration Stick" and Shift-Click on two of your calibration marks. Enter the distance between the two points to calibrate the video.
q) Repeat above for your other two calibration points.
r) You should now end up with a table of t(s), x(m) and y(m) for your ball as well as graphs of $y(t)$ and $x(t)$ as shown below.
s) You can also get tracker to display velocity and acceleration vectors which you can scale to look pretty.

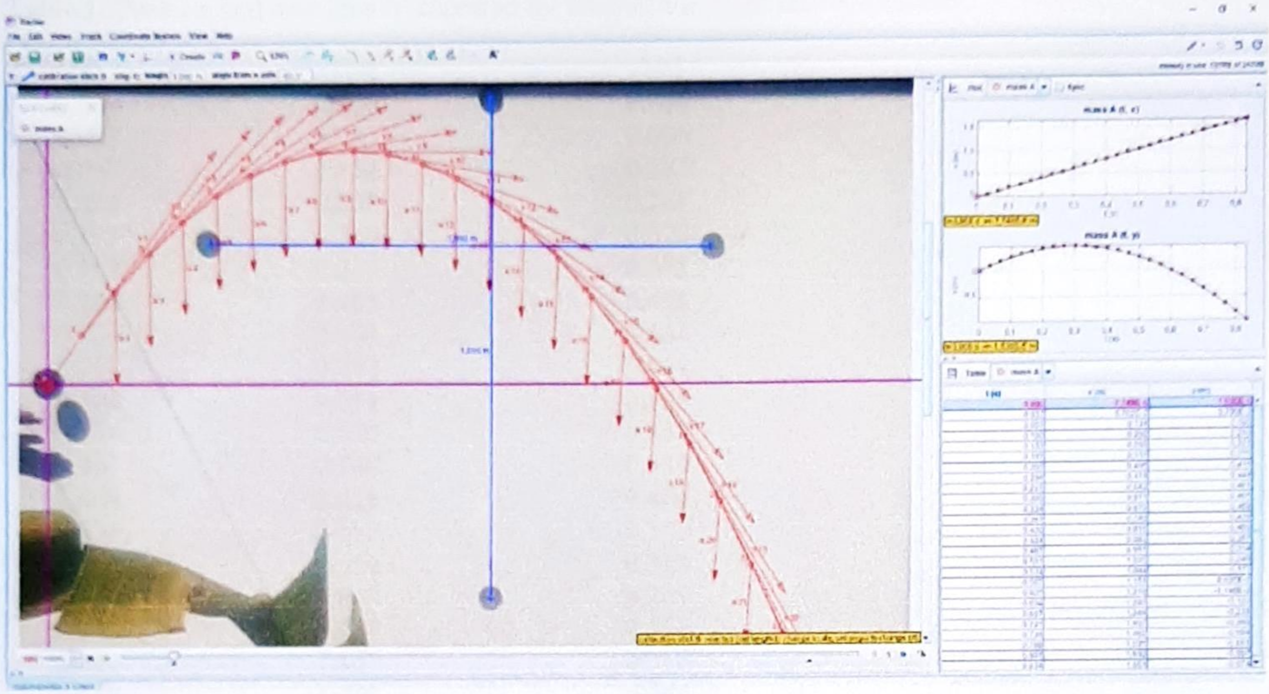


Figure 1: Screen shot of Tracker after capturing the whole motion. The blue lines are the "calibration sticks" that calibrate the distance between the markers. The numbers next to the points indicate the frame number, and v and a indicate the calculated velocity and acceleration vectors. On the right are graphs of $x(t)$ and $y(t)$ as well as a table with the (calibrated) raw data.

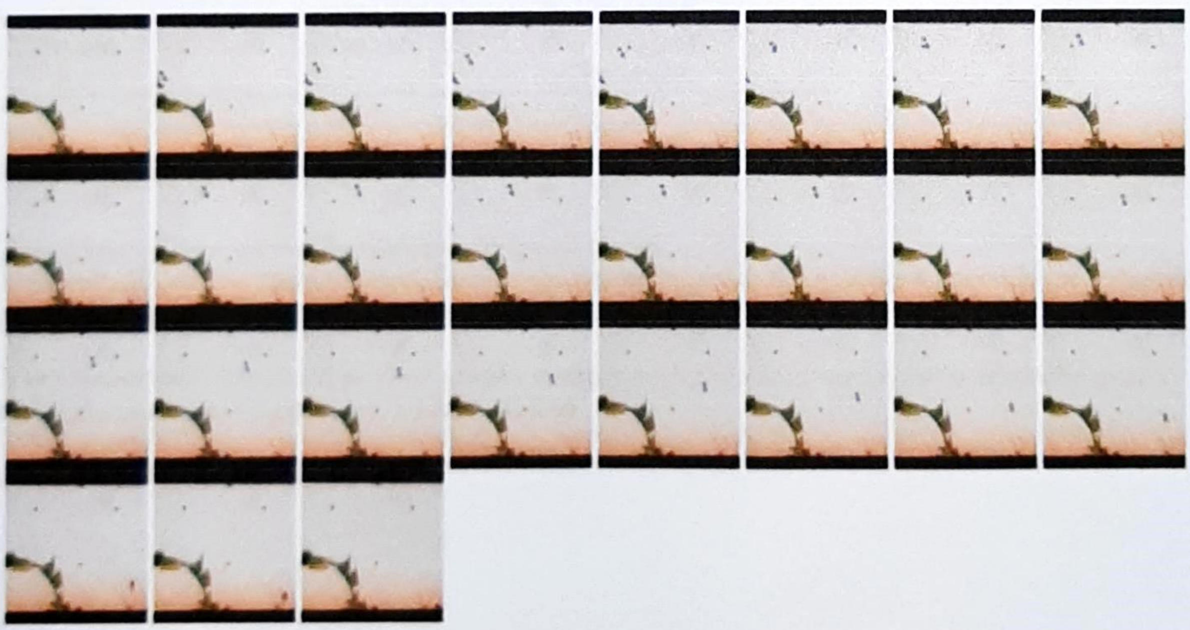


Figure 2: The 28 frames from which the data was collected at a frame rate of 30 frames per second. Note the four dots in the shape of a cross (length of arms 1.00 m) that were used to calibrate the length scale. The ball travels from the left to the right, and is most easily identified by the shadow it casts on the wall. In principle, one could use a ruler to find the x- and y-coordinates of the ball and apply the scaling to get to the data below.

PART B: Data analysis

Below is the captured data.

Table 1: Position and time data as captured by Tracker for a ball that was thrown.

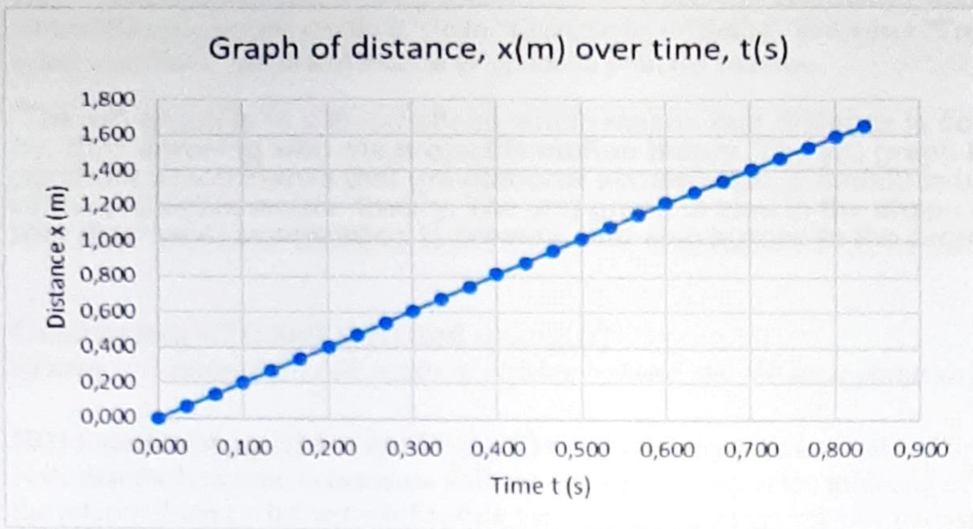
t (s)	x (m)	y (m)
0.000	**0.000**	**0.000**
0.033	0.067	0.098
0.067	0.134	0.185
0.100	**0.202**	**0.260**
0.133	0.269	0.323
0.167	0.337	0.375
0.200	**0.405**	**0.416**
0.234	0.474	0.444
0.267	0.543	0.461
0.300	**0.611**	**0.465**
0.334	0.679	0.458
0.367	0.748	0.439
0.400	**0.816**	**0.409**
0.434	0.883	0.367
0.467	0.951	0.314
0.500	**1.018**	**0.249**
0.534	1.084	0.173
0.567	1.151	0.086
0.600	**1.216**	**-0.011**
0.634	1.280	-0.120
0.667	1.344	-0.238
0.701	1.407	-0.366
0.734	1.469	-0.504
0.768	1.531	-0.651
0.801	1.592	-0.807
0.834	1.651	-0.974

[B1] Plot graphs of $x(t)$, $y(t)$ and $y(x)$.

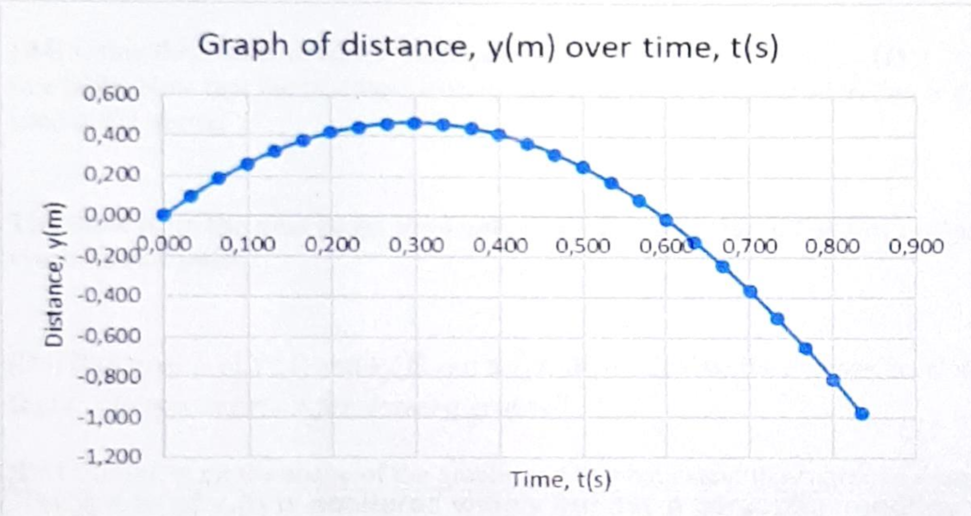
Computer: If you use a computer, please use all points.
By hand: If you plot them by hand, just use the points from 0 to 0.6 s, in 0.1 s intervals (as indicated in bold). Graph paper is included at the end of this document.

Very important: Ensure that your graphs comply with the requirements of a scientific graph as we discussed in Experiment 1 and 2 (GRA)

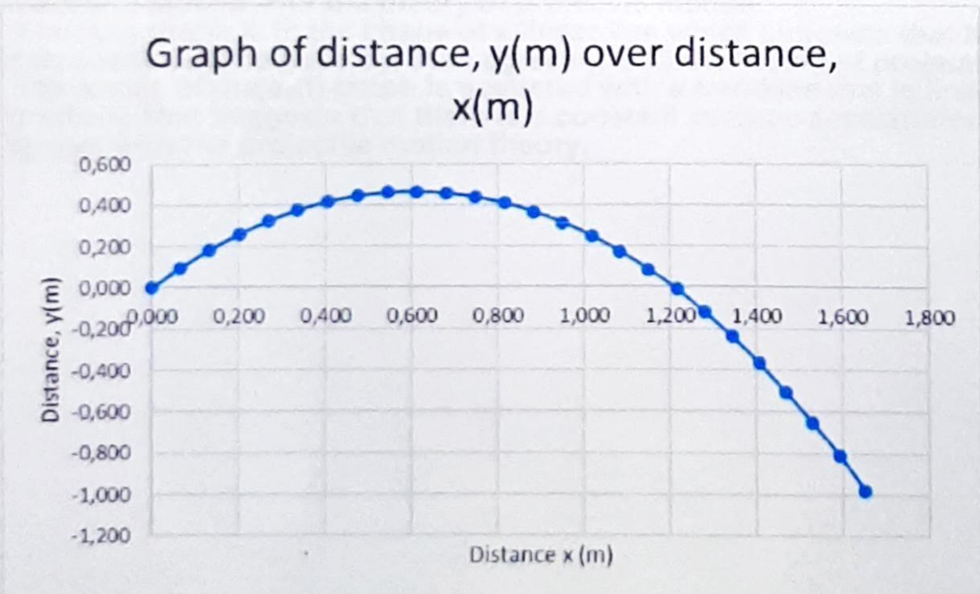
GRAPHS



Graph of distance, x(m) over time, t(s)
Distance x (m)
1,800
1,600
1,400
1,200
1,000
0,800
0,600
0,400
0,200
0,000
0,000
0,100
0,200
0,300
0,400
0,500
0,600
0,700
0,800
0,900
Time t (s)



Graph of distance, y(m) over time, t(s)
Distance, y(m)
0,600
0,400
0,200
0,000
-0,200
-0,400
-0,600
-0,800
-1,000
-1,200
0,000
0,100
0,200
0,300
0,400
0,500
0,600
0,700
0,800
0,900
Time, t(s)



Graph of distance, y(m) over distance, x(m)
Distance, y(m)
0,600
0,400
0,200
0,000
-0,200
-0,400
-0,600
-0,800
-1,000
-1,200
0,000
0,200
0,400
0,600
0,800
1,000
1,200
1,400
1,600
1,800
Distance x (m)

[B2] Comment on the shape of the graphs and to what extent they agree or disagree with the theory. *Hint*: Fit a quadratic curve to the graph of $y(t)$ and compare its coefficients with the expected experimental values. (Google graphs can do it. Go to "Chart Style"->"Series" and select "Trend Line". There you can select a quadratic graph and there is an option to print the equation.

The x(t) graph is in a linear shape which means that distance is continuous as time goes by, thus agreeing with the projectile motion theory. The y(t) graph is in the shape of a parabola which shows that gravitational acceleration (9.8m/s2) is involved, also an aspect of the projectile motion theory. The y(x) graph is also in the shape of a parabola that shows that downward acceleration is present, and also agrees to the projectile motion theory.

Calculating $vx(t)$ and $vy(t)$ and $ay,avg,(t)$

(If using a computer, plot all points, if plotting by hand, use the same points as previously.)

[B3] Calculate $v_{x,avg},(t) = x/\ t$ and $v_{y,avg},(t) = y/\ t$ for every time interval and present your results in a table. Note that the best time to associate with the average velocity is the midpoint of the interval. So, if you use the interval from $t = 0.4$ to $t = 0.5$ s, then the calculated average velocity corresponds best with the instantaneous velocity at $t = 0.45$ s.

[B4] Using the results obtained in the previous section, calculate $a_{y,avg},(t) = v_{y,avg}/\ t$, present your results in a table. Note that the best time to associate with the average acceleration is the midpoint of the intervals used in B3 above.

The table is on the next page. Remember to label the column headings using the correct format, with symbols and units.

[B4] Plot graphs of $v_x(t)$ and $v_y(t)$ and $a_y(t)$. Also calculate the average acceleration of the ball during its flight. *(Remember rules for drawing graphs!)*

[B5] Comment on the shape of the graphs and to what extent they agree or disagree with the theory.

The graph of $v_x(t)$ is scattered widely but has a parabolic trendline which suggests that the average velocity decreases over the midpoint intervals of time due to air resistance. This therefore agrees with the theory of projectile motion.
The $v_y(t)$ graph is in the shape of a linear line which suggests that the average velocity decreases as time goes by, thus agreeing with the theory of projectile motion.
The shape of the $a_y(t)$ graph is scattered with a trendline that is linear and has a small gradient, that suggests that there is a constant average acceleration at some point. This agrees with the projectile motion theory.

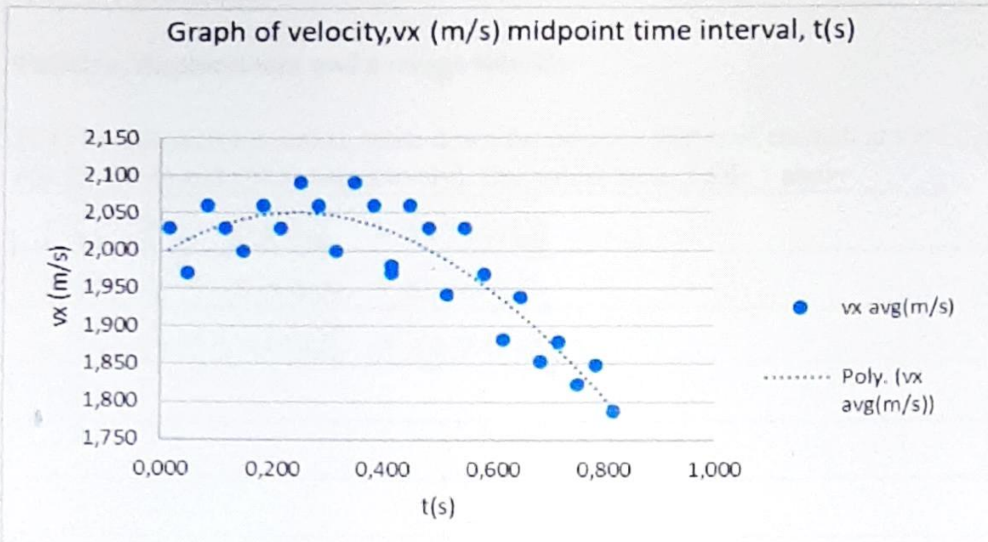
TABLE 2

Title: Table plotting Time t(s), Horizontal distance x(m), Vertical distance y(m), Midpoint of time intervals: mid t1(s); mid t2(s), Average velocity V_x(m/s), Average velocity V_y(m/s), Average acceleration A_y(m/s2) of the projectile motion of a ball.

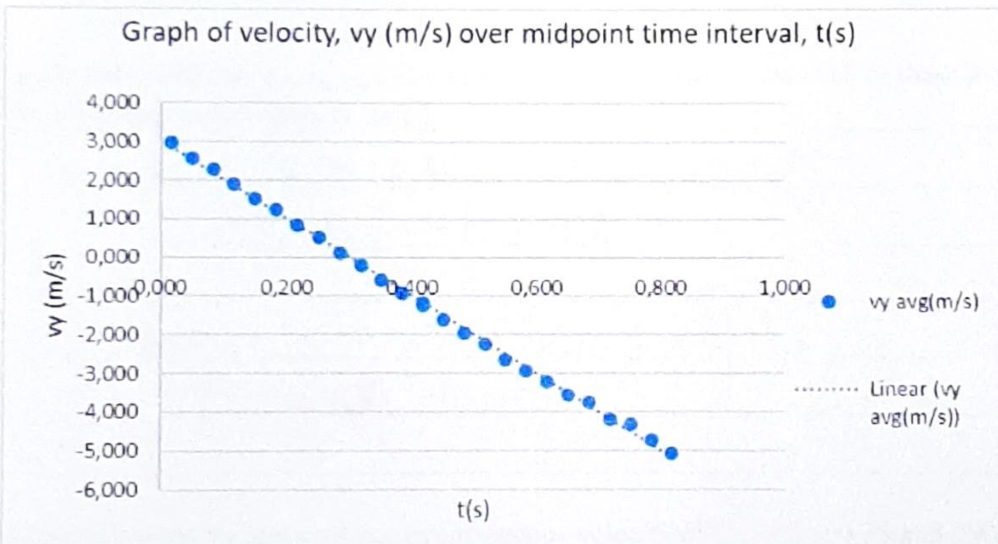
			Midpoint of time interval	*Average velocity*	*Average velocity*	*Midpoint of time interval*	*Average acceleration*
t (s)	*x* (m)	*y* (m)	mid t1(s)	V_x(m/s)	V_y(m/s)	mid t2(s)	A_y(m/s2)
0.000	**0.000**	**0.000**	0,017	2,030	2,970	0,033	-12,265
0.033	0.067	0.098	0,050	1,971	2,559	0,067	-8,604
0.067	0.134	0.185	0,084	2,061	2,273	0,100	-10,936
0.100	**0.202**	**0.260**	0,117	2,030	1,909	0,133	-11,334
0.133	0.269	0.323	0,150	2,000	1,529	0,167	-8,567
0.167	0.337	0.375	0,184	2,061	1,242	0,200	-12,504
0.200	**0.405**	**0.416**	0,217	2,029	0,824	0,234	-9,275
0.234	0.474	0.444	0,251	2,091	0,515	0,267	-11,848
0.267	0.543	0.461	0,284	2,061	0,121	0,300	-9,764
0.300	**0.611**	**0.465**	0,317	2,000	-0,206	0,334	-11,124
0.334	0.679	0.458	0,351	2,091	-0,576	0,367	-10,025
0.367	0.748	0.439	0,384	2,061	-0,909	0,400	-9,737
0.400	**0.816**	**0.409**	0,417	1,971	-1,235	0,434	-11,151
0.434	0.883	0.367	0,451	2,061	-1,606	0,467	-10,936
0.467	0.951	0.314	0,484	2,030	-1,970	0,500	-7,928
0.500	**1.018**	**0.249**	0,517	1,941	-2,235	0,534	-12,062
0.534	1.084	0.173	0,551	2,030	-2,636	0,567	-9,114
0.567	1.151	0.086	0,584	1,970	-2,939	0,600	-7,955
0.600	**1.216**	**-0.011**	0,617	1,882	-3,206	0,634	-11,041
0.634	1.280	-0.120	0,651	1,939	-3,576	0,667	-5,640
0.667	1.344	-0.238	0,684	1,853	-3,765	0,701	-12,451
0.701	1.407	-0.366	0,718	1,879	-4,182	0,734	-4,230
0.734	1.469	-0.504	0,751	1,824	-4,324	0,768	-12,143
0.768	1.531	-0.651	0,785	1,848	-4,727	0,801	1,814
0.801	1.592	-0.807	0,818	1,788	-5,061	0,617	-9,524
0.834	1.651	-0.974	0,417	1,980	-1,168	0,209	-5,601

Average acceleration = -9,383 m/s2

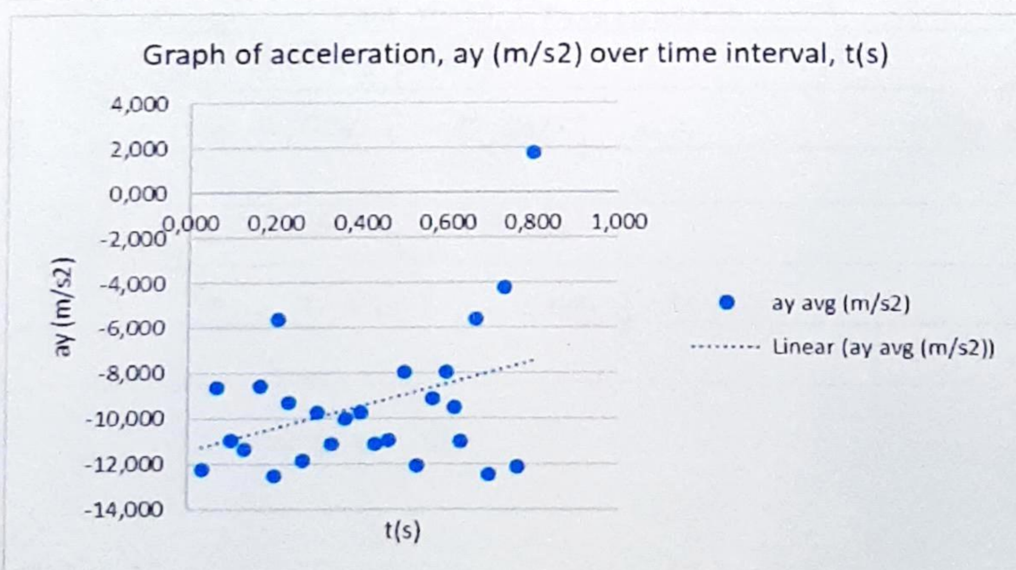
GRAPHS



Graph of velocity,vx (m/s) midpoint time interval, t(s)
2,150
2,100
2,050
2,000
1,950
1,900
1,850
1,800
1,750
vx (m/s)
0,000
0,200
0,400
0,600
0,800
1,000
t(s)
vx avg(m/s)
Poly. (vx avg(m/s))



Graph of velocity, vy (m/s) over midpoint time interval, t(s)
4,000
3,000
2,000
1,000
0,000
-1,000
-2,000
-3,000
-4,000
-5,000
-6,000
vy (m/s)
0,000
0,200
0,400
0,600
0,800
1,000
t(s)
vy avg(m/s)
Linear (vy avg(m/s))



Graph of acceleration, ay (m/s2) over time interval, t(s)
4,000
2,000
0,000
-2,000
-4,000
-6,000
-8,000
-10,000
-12,000
-14,000
ay (m/s2)
0,000
0,200
0,400
0,600
0,800
1,000
t(s)
ay avg (m/s2)
Linear (ay avg (m/s2))

PART C: Vectors:

Position, displacement and average velocity.

[C1] In unit vector notation, write down the position vector of the ball at t = 0.3, 0.4 and 0.5 s. [Call them **r**(0.3), **r**(0.4) and **r**(0.5) respectively]. Use values from Table 1 above.

$r(0{,}3) = (0{,}611\,m)\hat{i} + (0{,}465\,m)\hat{j}$

$r(0{,}4) = (0{,}816\,m)\hat{i} + (0{,}409\,m)\hat{j}$

$r(0{,}5) = (1{,}018\,m)\hat{i} + (0{,}249\,m)\hat{j}$

[C2] Now calculate $\mathbf{r}_{t=0.3 \text{ to } 0.4\text{ s}}$ and $\mathbf{r}_{t=0.4 \text{ to } 0.5\text{ s}}$. Again, write each of these in unit vector notation, and do not forget vector lines or units!

$r_{0{,}3 \text{ to } 0{,}4} = (0{,}816 - 0{,}611)\hat{i} + (0{,}409 - 0{,}465)\hat{j}$

$= (0{,}205\,m)\hat{i} - (0{,}056\,m)\hat{j}$

$r_{0{,}4 \text{ to } 0{,}5} = (1{,}018 - 0{,}816)\hat{i} + (0{,}249 - 0{,}409)\hat{j}$

$= (0{,}202\,m)\hat{i} - (0{,}160\,m)\hat{j}$

[C3] Calculate the approximate instantaneous velocity of the ball at 0.35 and 0.45 s using $\mathbf{v}(0.35) \approx \mathbf{r}_{t=0.3 \text{ to } 0.4\text{ s}}/t$ and $\mathbf{v}(0.45) \approx \mathbf{r}_{t=0.4 \text{ to } 0.5\text{ s}}/t$.

$V(0{,}35) = \dfrac{0{,}205\hat{i} - 0{,}056\hat{j}}{0{,}1}$

$= 2{,}050\hat{i} - 0{,}560\hat{j}$ m.s

$\Delta t = 0{,}45 - 0{,}35$

$= 0{,}100$ s

$V(0{,}45) = \dfrac{0{,}202\hat{i} - 0{,}160\hat{j}}{0{,}1}$

$= 2{,}020\hat{i} - 1{,}600\hat{j}$ m.s

[C4] Using your answer in the previous question, calculate the speed of the ball at time t = 0.35 and t = 0.45s (i.e. **v**(0.35) and **v**(0.45)), as well as the angles that the direction that the ball travels in makes with the positive x-axis at these two instants. Call them $\theta_{t=0.35\,s}$ and $\theta_{t=0.45\,s}$. Give the angle relative to the horizontal, with upwards positive, in degrees.

$|V_{0,45}| = \sqrt{(2{,}020)^2 + (1{,}600)^2}$

$= 2{,}577 \, m \cdot s$

$\tan\theta = \frac{1{,}600}{2{,}020}$

$\theta = -38{,}382^\circ + 360^\circ$

$\therefore \theta_{t=0,45} = 321{,}618^\circ$

$|V_{0,35}| = \sqrt{(2{,}050)^2 + (0{,}560)^2}$

$= 2{,}125 \, m \cdot s$

$\tan\theta = \frac{0{,}560}{2{,}050}$

$\theta = 15{,}279^\circ$

$\therefore \theta_{t=0,35} = 360 - 15{,}279^\circ$

$= 344{,}72^\circ$

[C5] Comment on your answers in [C3] and [C4] and how they correspond to theory.

Horizontal velocity decreases over time, which agrees to the theory (air resistance plays a big role). The vertical velocity also decreases over time because of gravitational force and corresponds to the theory. The effect of air resistance and gravity can be seen by the ball losing speed and the ~~de~~ decreasing of the angle between the projectile and the horizontal plane.

[C6] Using your previous results, calculate the approximate acceleration at time t = 0.4 s, **a**(0.4). Also calculate its magnitude and the angle that it makes with the horizontal.

$a(0{,}4) = \frac{V(0{,}45) - V(0{,}35)}{\Delta t}$

$= \frac{(2{,}202\,\hat{i} - 0{,}160\,\hat{j}) - (0{,}205\,\hat{i} - \cancel{0{,}056}\,56\,\hat{j})}{0{,}1}$

$= \frac{0{,}003\,\hat{i} - \cancel{0{,}104}\,1{,}04\,\hat{j}}{0{,}1}$

$= (0{,}03 \, m \cdot s^2)\,\hat{i} - (\cancel{1{,}04}\,10{,}4 \, m \cdot s^2)\,\hat{j}$

$\tan\theta = \frac{y}{x}$

$\theta = \tan^{-1}\left(\frac{10{,}4}{0{,}03}\right)$

$= 89{,}835^\circ$

$|a_{(0,4)}| = \sqrt{(\cancel{1{,}04}\,0{,}03)^2 + (-\cancel{1{,}04}\,10{,}4)^2}$

$= \cancel{1{,}040} \, m \cdot s^2$

$10{,}400$

$\Delta t = 0{,}45 - 0{,}35$

$= 0{,}1$

[C7] Comment on your answer in [C6] and how well it corresponds to theory.

The magnitude that is calculated is this experiment is close to gravitational acceleration (9.8m/s).
This is proof that the results are measured and calculated very accurately. The acceleration v_y, is almost completely vertical because of the forces like gravitational force and air resistance being vertical forces.

PART D: Discussion:

Are the results obtained in this experiment 100% accurate? List the aspects that you think could contribute to answers deviating from the ideal.

No the results obtained in this experiment are not 100% accurate and this could be due to air resistance, timing errors and human errors